Anshul **Nayak**

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Research Interests

Vision-language critic, Multi-agent reinforcement learning, Trajectory Optimization (MPC, MPPI, LQR, PID), Path Planning (A*, RRT, PRRT), Uncertainty quantification, Deep Learning (LSTM, CNN, Transformers), NN Pipeline (Data creation, training, hyperparameter tuning, evaluation), navigation and obstacle avoidance, Embedded hardware, Vehicle kinematics and control (wheeled robot, arms), Sensor integration and data collection (IMU, GPS, Lidar, Camera, Stereo Vision), Cooperative perception and pose estimation, Calibration, State Estimation, Sensor Fusion and Probabilistic Filters (KF, EKF), Localization and Mapping, Computer vision (object detection and tracking, depth estimation), Lidar Processing (PCL, Open3D)

Open to opportunities in the field of deep learning, robotics, control and autonomous systems.

Technical Skills

Languages	Python, C++ MATLAB
Real Time systems	Arduino, Jetson
Simulation Software	Gazebo, IsaacSim, CARLA
Frameworks	PyTorch, CasADi, Tensorflow, ROS/ROS2, Nav2, OpenCV, Linux, Docker, git
Relevant coursework	Deep Learning, Reinforcement Learning, Model Predictive Control, Optimisation, Non-Linear Systems Theory

Education

Virginia Tech *Blacksburg, Virginia*
PH.D. IN MECHANICAL ENGINEERING *Jan. 2021 - Dec. 2024*
THESIS: "COOPERATIVE PREDICTION AND PLANNING UNDER UNCERTAINTY FOR AUTONOMOUS ROBOTS."
- Adviser: Dr. Azim Eskandarian

M.S IN MECHANICAL ENGINEERING *Aug. 2018 - Dec. 2020*
THESIS: "EFFECT OF KINEMATICS AND CAUDAL FIN PROPERTIES ON PERFORMANCE OF A FREELY-SWIMMING FIN"
- Adviser: Dr. Hodjat Pendar

NIT Rourkela
B.TECH IN MECHANICAL ENGINEERING *May. 2011 - June. 2015*
- Adviser: Dr. Soumya Gangopadhyay

Work Experience

Clemson University *Clemson, South Carolina*
POSTDOCTORAL SCHOLAR *Nov. 2025 - Current*
- Developed a **belief-aware VLM** with memory retrieval for long-horizon intent inference and human-like reasoning.
- Proposed **MA-VLCM** that replaces learned critic with a pretrained VLM for sample efficiency and zero-shot generalizable multi-agent policies.

Virginia Tech Transportation Institute *Blacksburg, Virginia*
RESEARCH ASSOCIATE *Jan. 2025 - Oct. 2025*
- Developed a ROS2-enabled Jetbot with onboard depth sensors in Isaacsim for planning and obstacle avoidance.
- Enhanced long-range detection using camera and range sensors for autonomous emergency braking.
- Simulated rainy conditions using monocular depth estimation and physics-based particles to study sensor degradation.

Honda Research Institute *San Jose, California*
RESEARCH INTERN, ADVISER: DR. SONGPO LI *Feb. 2024 - May. 2024*
- Developed a 7 DOF human arm model using **Model Predictive control (MPC)** **for coordinated reaching and orientation** .
- Optimized cost function weights to generate human-like trajectories and validated with actual upper-limb data.

Virginia Tech *Blacksburg, Virginia*
RESEARCH ASSISTANT, ADVISER: DR. AZIM ESKANDARIAN *Jan. 2021 - Dec. 2024*
- Developed an **uncertainty-aware social navigation planner** using chance-constrained optimal control and Control Barrier Functions (CBFs).
- Developed a **vision-based obstacle avoidance planner** by learning cost function maps as potential fields.
- Developed a seq2seq Kalman filter surrogate to capture and propagate sensing uncertainty via **deep ensembles** for robust trajectory prediction.
- Created a Bayesian inference model with **Monte Carlo Dropout** for trajectory prediction, improved accuracy by 15% over deterministic models.
- Designed a **cooperative prediction algorithm** to forecast trajectories of occluded objects in GPS-denied environments.
- Proposed a **probabilistic RRT (PRRT)** planner using Gaussian Processes, enhancing proactive collision avoidance.
- Built and deployed the **autonomous X-car** platform at Virginia Tech using ROS and CARMA, with SLAM, V2V communication, waypoint generation, and data processing capabilities.

Publications

PEER-REVIEWED JOURNALS

- **Nayak A** and Eskandarian A, Prediction uncertainty-aware planning using deep ensembles and trajectory optimization. (under review)
- **Nayak A.**, Eskandarian A, Doerzaph Z., and Ghorai P. *"Pedestrian Trajectory Forecasting Using Deep Ensembles Under Sensing Uncertainty."* IEEE Transactions on Intelligent Transportation Systems (2024).
- **Nayak A.** , and Eskandarian A., "Cooperative Probabilistic Trajectory Prediction under Occlusion," in IEEE Transactions on Intelligent Vehicles, doi: 10.1109/TIV.2024.3365651.
- **Nayak A.**, Eskandarian A. and Doerzaph Z., *"Uncertainty Estimation of Pedestrian Future Trajectory Using Bayesian Approximation,"* in IEEE Open Journal of Intelligent Transportation Systems, vol. 3, pp. 617-630, 2022, doi: 10.1109/OJITS.2022.3205504.
- G. Mehr, Prasenjit Ghorai, Ce Zhang, **Anshul Nayak**, Darshit Patel, Shathushan Sivashangaran, Azim Eskandarian, *"X-CAR: An Experimental Vehicle Platform for Connected Autonomy Research,"* in IEEE Intelligent Transportation Systems Magazine, doi: 10.1109/MITS.2022.3168801.
- Wu, X., **Nayak, A.**, and Eskandarian, A., *"Motion Planning of Autonomous Vehicles under Dynamic Traffic Environment in Intersections Using Probabilistic Rapidly Exploring Random Tree,"* SAE Intl. J CAV 4(4):383-399, 2021, https://doi.org/10.4271/12-04-04-0029.
- **Nayak, A.**, S. Gangopadhyay, and D. K. Sahoo. *"Modelling, simulation and experimental investigation for generating 'I'shaped contour on Inconel 825 using electrochemical machining."* Journal of Manufacturing Processes 23 (2016): 269-277.
- **Nayak A.**, Masroor E. and Pendar H. *"Performance comparison of a tethered and freely-swimming fin" (Under Review)* in Bioinspiration and Biomimetics

PEER-REVIEWED CONFERENCE PROCEEDINGS

- **Nayak, A**, Eskandarian, A, Ghorai, P and Doerzaph, Z. *"A Comparative Study on Feature Descriptors for Relative Pose Estimation in Connected Vehicles."* Proceedings of the ASME 2021 International Mechanical Engineering Congress and Exposition. Volume 7B Dynamics, Vibration, and Control. Virtual, Online. November 1–5, 2021. V07BT07A022. ASME. https://doi.org/10.1115/IMECE2021-70693
- Ghorai, P, Eskandarian, A, **Nayak, A**, and Doerzaph, Z. *"Relative Pose Estimation for Cooperative Vehicles and Tracking of Multiple Dynamic Objects."* Proceedings of the ASME 2021 International Mechanical Engineering Congress and Exposition. Volume 7B Dynamics, Vibration, and Control.Virtual, Online. November 1–5, 2021. V07BT07A017. ASME. https://doi.org/10.1115/IMECE2021-69651
- **Nayak A.**, Hodjat Pendar. *"Effect of kinematics and physical properties of caudal fin on performance of a robotic fish."* In: APS March Meeting Abstracts, 2021:B14. 011, March 15–19, 2021. Virtual conference.

Past Experience

Virginia Tech

GRADUATE STUDENT RESEARCHER, ADVISER: DR. HODJAT PENDAR *Aug. 2018 - Dec. 2020*

- Designed a **robotic fish to investigate the effect of caudal fin** properties like fin flexibility, shape and thickness on propulsion performance.
- Developed a **numerical model to simulate fish dynamics**, comparing thrust and efficiency between freely-swimming and tethered fish.

Bajaj Auto Limited *Pune , India*

GRADUATE TRAINEE ENGINEER *Jul. 2015 - Jul. 2018*

- Co-Lead a team on **Multi-Body Dynamics simulation using MSC ADAMS** for chassis subsystems.
- Lead a project towards **improving rolling resistance of tires** and **increased mileage** of commercial electric vehicles by 3%.
- Optimised performance of engines using Ricardo Wave and star-CCM.

Indian Institute of Science *Bangalore, India*

SUMMER RESEARCH FELLOW *Apr. 2014 - Jul. 2014*

- Contributed towards experimental **investigation of swirl flows** inside a compact combustor.
- Performed **numerical simulation of swirl patterns** to validate experiment using **ANSYS CFD** and reducing overall experimental dependency.

Honors & Awards

2024	**Fellowship**, Prestigious **Pratt Fellowship**, Mechanical Engineering Department	*Blacksburg, U.S.A*
2014	**Fellowship**, Indian Academy of Sciences, IASc-INSA-NASI, Summer Research Fellow	*IISc, Bangalore*
2012	**Finalist**, Mahindra **Baja SAE**, Indian student chapter	*Pithampur, India*

Services

Reviewer	IEEE T-ITS, IEEE T-IV, IEEE ITSC
External Reviewer	IEEE-Intelligent vehicle symposium, IEEE IV 2022